
3AL Tutorial 2 2024 Solutions

1. Let G be a group of order 105. List all possible orders of elements of G .

By Lagrange's Theorem, if H is a subgroup of G , the order of H must divide $|G| = 105$. So, we need to find the divisors of 105. They are 1, 3, 5, 7, 15, 21, 35 and 105. So, these are the possible orders of subgroups of G .

2. Let p and q be distinct primes and G a group with order pq . Prove that every proper subgroup of G is cyclic.

NOTE: We call a subgroup "proper" if it does not equal the group itself.

Let H be a proper subgroup of G . Then $|H| < |G| = pq$ and so, by Lagrange's Theorem, $|H| = 1, p$ or q . If $|H| = 1$, then $H = \langle 1 \rangle$. If $|H| = p$, then H is a group of prime order and so it is cyclic. Similarly if $|H| = q$.

3. Let G be a group with subgroups H and K . Prove the following statements:

- (a) $H \triangleleft G$ if and only if H is the kernel of a homomorphism.
- (b) If H and K are normal in G , then $H \cap K \triangleleft G$.
- (c) If $H \triangleleft G$, then $H \cap K \triangleleft K$.
- (d) If $H \triangleleft G$ and $H \subseteq K$, then $H \triangleleft K$.

(a) Suppose H is a normal subgroup of G . Consider the natural projection homomorphism $\pi : G \rightarrow G/H$ defined by $\pi(g) = gH$. Notice that $\pi(g) = 1_{G/H} = H$ if and only if $g \in H$. That is, $\ker \pi = H$.

NOTE: If you haven't already proved that φ is a homomorphism, please do so now.

Now suppose $H = \ker \varphi$ for some homomorphism $\varphi : G \rightarrow G'$. We need to show

$H \triangleleft G$. Let $g \in G$, we will show $g \ker \varphi g^{-1} \subseteq \ker \varphi$. That is, for all $x \in \ker \varphi$, we have $g x g^{-1} \in \ker \varphi$. In particular, $\varphi(g x g^{-1}) = 1_{G'}$. Note that

$$\varphi(g x g^{-1}) = \varphi(g) \varphi(x) \varphi(g)^{-1} = \varphi(g) 1_{G'} \varphi(g)^{-1} = 1_{G'}.$$

So, $H = \ker \varphi \triangleleft G$.

(b) Let $g \in G$, we will show $g(H \cap K)g^{-1} \subseteq H \cap K$. Take $x \in H \cap K$. Consider $g x g^{-1}$. Since $H \triangleleft G$ and $x \in H$, we have $g x g^{-1} \in H$. Similarly, since $K \triangleleft G$ and $x \in K$, we have $g x g^{-1} \in K$. Thus, $g x g^{-1} \in H \cap K$. So, we can conclude that $g(H \cap K)g^{-1} \subseteq H \cap K$ for all $g \in G$ and so $H \cap K \triangleleft G$.

QUESTION: Is the statement true if only one of H and K is normal in G ?

(c) Let $k \in K$, we will show that $k(H \cap K)k^{-1} \subseteq H \cap K$. Take $x \in H \cap K$. Since $H \triangleleft G$, we have $g x g^{-1} \in H$ for all $g \in G$. This means $k x k^{-1} \in H$. Now, since $x \in H \cap K$, we have $x \in K$. Therefore $k x k^{-1} \in K$ as K is a subgroup. Thus $x \in H \cap K$ and so $k(H \cap K)k^{-1} \subseteq H \cap K$. So $H \cap K \triangleleft K$.

(d) If $H \subseteq K$, then $H \cap K = H$ and so $H \triangleleft K$ by the previous part.

4. Let G be a group with subgroup H . Prove that if $|G : H| = 2$, then $H \triangleleft G$.

Let $|G : H| = 2$. Then there are two left cosets of H . One of the cosets must be H itself. Since the left cosets of a subgroup partition G , all elements of G that are not in H must appear in the other left coset. So, the other left coset can be written as $G \setminus H = \{g \in G : g \notin H\}$. Through similar reasoning, the only right cosets of H are H and $G \setminus H$.

To show that $H \triangleleft G$, we will show that $gH = Hg$ for all $g \in G$. If $g \in H$, then $gH = H = Hg$. On the other hand, if $g \notin H$, then $gH = G \setminus H$ and $Hg = G \setminus H$. Thus, $gH = Hg$ for all $g \in G$. So, $H \triangleleft G$.

5. Let G be a group. Prove the following:

- (a) If $G/Z(G)$ is cyclic, then G is abelian.

HINT: Let the generator of $G/Z(G)$ be denoted by gH (where $H = Z(G)$). Show that every element of G can be written as $g^n z$ for $n \in \mathbb{Z}$ and $z \in Z(G)$.

- (b) If G has order pq (for primes p and q , not necessarily distinct), then either G is abelian or $Z(G) = \{1\}$.

(a) Let $H = Z(G)$. Suppose $G/Z(G)$ is cyclic. Let $gH \in G/Z(G)$ represent the generator of $G/Z(G)$. Then let a be an arbitrary element of G . Then $aH = (gH)^n$ for some $n \in \mathbb{Z}$. That is, $aH = g^n H$. By coset properties, this means $a \in g^n H$ and so $a = g^n h$ for some $h \in H$.

Take $a, b \in G$. In order to show that G is abelian, we will show that $ab = ba$. From above, we can write $a = g^n h$ and $b = g^{n'} h'$ for some $n, n' \in \mathbb{Z}$ and $h, h' \in H = Z(G)$. Thus, $ab = g^n h g^{n'} h'$. However, h and h' commute with all elements, so we have

$$ab = g^n h g^{n'} h' = g^n g^{n'} h h' = g^{n'} g^n h' h = g^{n'} h' g^n h = ba.$$

Therefore, all elements in G commute. Thus G is abelian.

(b) Since $Z(G)$ is a subgroup of a finite group G , Lagrange's Theorem tells us that either $|Z(G)| = 1, p, q$ or pq . If $|Z(G)| = p$, then $|G/Z(G)| = q$ and so $G/Z(G)$ is cyclic because it has prime order. Thus, by (a), G is abelian. Similarly if $|Z(G)| = q$. If $|Z(G)| = pq$, we must have $G = Z(G)$ and so, again, G is abelian. Of course, if $|Z(G)| = 1$, then $Z(G) = \{1\}$.

6. Let $G = \mathbb{Z}_4 \times \mathbb{Z}_4$ and $H = \{(0, 0), (2, 0), (0, 2), (2, 2)\}$ (here we drop the equivalence class notation $[\]$ for ease of notation). Is G/H isomorphic to $\mathbb{Z}_4$ or K_4 ?

First we need to determine the elements of G/H so we can understand whether it has a cyclic structure (i.e. is isomorphic to $\mathbb{Z}_4$) or not (i.e. is isomorphic to K_4).

The elements of G/H are the cosets of H in G :

$$\begin{aligned} H &= \{(0, 0), (2, 0), (0, 2), (2, 2)\} \\ (1, 0) + H &= \{(1, 0), (3, 0), (1, 2), (3, 2)\} \\ (0, 1) + H &= \{(0, 1), (2, 1), (0, 3), (2, 3)\} \\ (1, 1) + H &= \{(1, 1), (3, 1), (1, 3), (3, 3)\}. \end{aligned}$$

Notice that these three sets partition all of $\mathbb{Z}_4 \times \mathbb{Z}_4$, so we know that these must be all of the cosets of H . We can see that G/H has order 4 thus must be isomorphic to either $\mathbb{Z}_4$ or K_4 . To check which, we will check if G/H is cyclic or not by examining the orders of the elements of G/H . Note that the order of H in G/H is 1. We also have

$$\begin{aligned} 2(1, 0) + H &= (2, 0) + H = H \\ 2(0, 1) + H &= (0, 2) + H = H \\ 2(1, 1) + H &= (2, 2) + H = H. \end{aligned}$$

(Notice that $(2, 0)$, $(0, 2)$ and $(2, 2)$ are all elements of H and so the cosets with those representatives must just be H). Now, we've seen that each element of G/H has order at most 2. So G/H does not have a generator and must be isomorphic to K_4 .

7. Let G be a group.

- (a) Prove that $\text{Aut}(G)$, the set of automorphisms $\varphi : G \rightarrow G$, is a group under composition.
- (b) Let $x \in G$. Prove that the map $\sigma_x : G \rightarrow G$ defined by $\sigma_x(g) = xgx^{-1}$ is an automorphism.
- (c) Let $x \in G$. Prove that $(\sigma_x)^{-1} = \sigma_{x^{-1}}$.
- (d) Prove that $\text{Inn}(G) = \{\sigma_x : x \in G\}$ is a subgroup of $\text{Aut}(G)$.
- (e) Find a group G such that $\text{Inn}(G) \cong \{e\}$ (the trivial group consisting only of an identity element).

(a) First note that composition of maps is always associative.

Now consider the identity map $\text{id}_G : G \rightarrow G$ defined by $\text{id}_G(g) = g$ for all $g \in G$. This is an automorphism (work through the details if you aren't comfortably with why—remember that an automorphism is a map $G \rightarrow G$ that is an *isomorphism*). We claim that id_G is the identity of $\text{Aut}(G)$. For any $\varphi \in \text{Aut}(G)$, we have $\text{id}_G(\varphi(g)) = \varphi(g) = \varphi(\text{id}_G(g))$. Thus $\text{id}_G \circ \varphi = \varphi \circ \text{id}_G$. So id_G is indeed the identity element of $\text{Aut}(G)$.

Now we show that $\text{Aut}(G)$ contains inverses. Let $\varphi \in \text{Aut}(G)$. Since $\varphi : G \rightarrow G$ is an isomorphism, it has an inverse $\varphi^{-1} : G \rightarrow G$. Recall that for a map to be an inverse, we must have $\varphi \circ \varphi^{-1} = 1_G = \varphi^{-1} \circ \varphi$. That is, φ^{-1} is the inverse of φ in the group $\text{Aut}(G)$.

Thus $\text{Aut}(G)$ is a group.

(b) Let $a, b \in G$. Then

$$\sigma_x(a)\sigma_x(b) = (xax^{-1})(xbx^{-1}) = x(ab)x^{-1} = \sigma_x(ab).$$

Thus σ_x is a homomorphism. We need to show that it is an isomorphism. Let $g \in G$, then $\sigma_x(x^{-1}gx) = g$ and so σ_x is an epimorphism. Suppose that $\sigma_x(a) = \sigma_x(b)$ for some $a, b \in G$. Then $xax^{-1} = xbx^{-1}$ and so $a = b$. Thus σ_x is a monomorphism. Finally, we can conclude that σ_x is an automorphism.

(c) We have $\sigma_x\sigma_x^{-1}(g) = x(x^{-1}gx)x^{-1} = g$ and $\sigma_x^{-1}\sigma_x(g) = x^{-1}(xgx^{-1})x = g$ for all $g \in G$. Thus the inverse of σ_x is $\sigma_{x^{-1}}$.

(d) From (b), it is clear that $\text{Inn}(G) \subseteq \text{Aut}(G)$. We now just need to use the Subgroup Test. Let $\sigma_x, \sigma_y \in \text{Inn}(G)$. We will show $\sigma_x\sigma_y^{-1} \in \text{Inn}(G)$.

(e) Any abelian group G will satisfy this since $\sigma_x = \text{id}_G$ for all $x \in G$. Thus $\text{Inn}(G) = \{\text{id}_G\} \cong \{e\}$.

Extra problems

(I) Let G be a group with $H \triangleleft G$. Prove that if $|G : H| = n$ (where $n \in \mathbb{Z}^+$), then $g^n \in H$ for all $g \in G$.

Let $g \in G$. Then $g^n H = (gH)^n$. We have $|G/H| = |G : H| = n$. Since the order of the factor group is n , all elements when raised to the power of n must equal the identity. Thus $(gH)^n = H$. That is $g^n H = H$ and so $g^n \in H$.

- (II) Show, via counterexample, that the image of a normal subgroup is not necessarily normal. That is, find groups G and G' with homomorphism $\varphi : G \rightarrow G'$ with $H \triangleleft G$ such that $\varphi(H) \not\triangleleft G'$.

Consider $\varphi : \mathbb{Z}_2 \rightarrow S_3$ defined by $\varphi([0]) = \varepsilon$ and $\varphi([1]) = \tau$. Work through the details yourself.